Best Practices for Writing Transparent and Reproducible Methods Sections in Geoscience Research

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Introduction

Methods sections are the foundation of scientific reproducibility, but many geoscience papers lack the detail needed for other researchers to verify or replicate studies. Clear, complete methods documentation is essential for advancing scientific knowledge and building trust in research findings. (LLM Memory)

The methods section serves as the blueprint that allows other scientists to understand, evaluate, and reproduce research findings. In geoscience research, where studies often involve complex field work, laboratory analyses, computational models, and data processing workflows, the methods section becomes particularly critical for scientific integrity (Model-Generated). Despite this importance, many published geoscience papers contain methods sections that lack sufficient detail for reproducibility, creating barriers to scientific verification and progress (Model-Generated).

Effective methods writing requires balancing completeness with clarity, ensuring that all essential information is provided while remaining accessible to readers with varying levels of expertise (Model-Generated). The goal is to create a comprehensive record that enables independent verification of results and facilitates future research building on the work (Model-Generated). This requires systematic attention to documentation standards, data accessibility, and technical implementation details that support transparent and reproducible science (Model-Generated).

Core Principles and Goals of Methods Sections

Methods sections serve three fundamental purposes: enabling verification of results and conclusions, assessing experimental validity, and facilitating reproducibility by other researchers. Effective methods writing requires plain, objective descriptions that follow logical organization and address the "who, what, when, where, how, and why" of the research. (3 sources)

The primary goals of methods sections center on scientific verification and reproducibility. A well-written methods section allows other scientists to verify results and conclusions, understand whether the experimental design appropriately addresses the research question, and build upon the work by reproducing results or testing alternative approaches (Zelst et al., 2021) (Zelst et al., 2022). This verification function is critical because the methods section provides the information by which a study's validity is judged (Zelst et al., 2021) (Kallet, 2004).

Effective methods writing follows several core principles. The description should be plain and simple, objective, and logically organized as a factual report of what was done in the study (Zelst et al., 2021) (Zelst et al., 2022). Unstructured and incomplete methods create barriers to understanding and can lead readers to question research validity (Zelst et al., 2021) (Zelst et al., 2022). A systematic approach addresses six key questions: who performed the experiment, what was done to answer the research question, when and where was the work undertaken, how was the experiment conducted and analyzed, and why were specific procedures chosen (Zelst et al., 2021) (Zelst et al., 2022).

Transparency in assumptions and methodology builds reader confidence. Model assumptions and limitations must be stated clearly and explicitly, as these may not be obvious to readers (Zelst et al., 2021) (Zelst et al., 2022). When assumptions are communicated clearly and consistently, reviewers and readers will find fewer flaws in the study (Zelst et al., 2021) (Zelst et al., 2022).

Evidence

(Zelst et al., 2021)

101 Geodynamic modelling: How to design, carry out, and interpret numerical studies

> *"The methods section is considered one of the most important parts of any scientific manuscript (Kallet, 2004). A good methods section allows other scientists to verify results and conclusions, understand whether the design of the experiment is relevant for the scientific question (validity), and to build on the work presented (reproducibility and replicability) by assessing alternative methods that might produce differing results. Thus, the major goals of the methods section are to verify the experiment layout and allow others to reproduce the results"*

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> *"After the model setup has been explained, the methods should contain a section describing the design or layout of the study in detail. What is being tested/varied? How many simulations were performed in terms of model and parameter space?"*

> *"Information should also be given on code and data availability"*

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How to Write the Methods Section of a Research Paper

> *"The methods section of a research paper provides the information by which a study's validity is judged. Therefore, it requires a clear and precise description of how an experiment was done, and the rationale for why specific experimental procedures were chosen. The methods section should describe what was done to answer the research question, describe how it was done, justify the experimental design, and explain how the results were analyzed. Scientific writing is direct and orderly. Therefore, the methods section structure should: describe the materials used in the study, explain how the materials were prepared for the study, describe the research protocol, explain how measurements were made and what calculations were performed, and state which statistical tests were done to analyze the data. Once all elements of the methods section are written, subsequent drafts should focus on how to present those elements as clearly and logically as possibly. The description of preparations, measurements, and the protocol should be organized chronologically. For clarity, when a large amount of detail must be presented, information should be presented in sub-sections according to topic. Material in each section should be organized by topic from most to least important."*

Structural Organization and Content Requirements

Methods sections should follow a logical hierarchical structure from study design overview through theoretical framework, experimental design, analysis methods, and computational details. Proper organization with clear subsections and chronological flow makes complex procedures accessible and verifiable. (4 sources)

Effective methods sections require systematic structural organization to present complex information clearly. The methods should begin with a brief outline describing the study design and main steps taken to answer the scientific question, following a logical progression from theoretical elements to numerical aspects to analysis and post-processing (Zelst et al., 2021). This hierarchical approach ensures readers can follow the research workflow from conception through execution.

The theoretical framework forms the foundation and should be described first, including mathematical and physical concepts, governing equations, constitutive equations, initial and boundary conditions, and other relevant theory describing the model setup (Zelst et al., 2021). Following the theoretical foundation, detailed description of the study design or layout should specify what is being tested or varied and how many simulations or experiments were performed across the model and parameter space (Zelst et al., 2021).

For complex studies requiring extensive detail, information should be organized in subsections according to topic, with material in each section arranged from most to least important (Zelst et al., 2021) (Kallet, 2004). The description of preparations, measurements, and protocols should follow chronological organization to maintain clarity and logical flow (Zelst et al., 2021) (Kallet, 2004).

Analysis, visualization, and post-processing techniques represent a commonly overlooked but essential component that should be explicitly documented. This includes specifying software used for visualization and data processing, statistical methods employed, and availability of processing scripts (Zelst et al., 2022). Clear structural organization with appropriate subsections and concise writing helps avoid jargon and ensures the methods section serves as an effective guide for reproducibility (Hillier et al., 2021).

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(Hillier et al., 2021)

Editorial: Geoscience communication – Planning to make it publishable

"When writing your paper, please endeavour to write clearly and concisely, avoid jargon (see e.g. Venhuizen et al., 2019) and include critical structural elements you will be familiar with (e.g. introduction, methods, results, discussion, conclusions)."

Documentation Requirements for Reproducibility

Reproducible geoscience research requires comprehensive documentation including complete metadata records, accessible datasets, computational details, analysis scripts, and explicit statements of model assumptions and limitations. (4 sources)

Essential documentation for reproducibility includes several key components that must be systematically addressed:

• **Complete metadata and data archiving**: All reviewed and interpreted datasets must be placed in permanent, trusted repositories with complete metadata records, as documentation forms the foundation of quality assurance and quality control (Medalie, 2020)

• **Computational environment specifications**: Authors should provide detailed information about where simulations were performed, number of cores used, average runtime, and whether results can be reproduced on desktop computers or require cluster access (Zelst et al., 2021) (Zelst et al., 2022)

• **Code and input file availability**: Documentation must include code availability statements, code verification details, input files, installation guides, and other data necessary to reproduce simulation results (Zelst et al., 2021) (Zelst et al., 2022)

• **Analysis and post-processing documentation**: Methods sections should explicitly describe visualization software, post-processing scripts, and statistical methods used for data analysis, with scripts made available for peer review and community engagement (Zelst et al., 2022)

• **Explicit model assumptions and limitations**: All model assumptions must be stated clearly and explicitly, as these may not be obvious to readers but are essential for proper interpretation and replication (Zelst et al., 2021) (Zelst et al., 2022)

• **Detailed supplementary materials**: Rather than lengthening main papers, adequate and detailed supplementary materials should be provided for readers aiming to reproduce work, supporting transparency of methods and minimum standards frameworks (Steventon et al., 2021)

Paper	Targeted Scientific Domain	Reproducibility Focus	Recommended Best Practices	Scope Of Reproducibility Issues Addressed	Methodology For Recommendations	Emphasis On Documentation
Considerations for incorporating quality control into water quality sampling strategies for the U.S. Geological Survey	water quality sampling strategies	Providing best practices for achieving reproducibility.	Recommendations for quality control	quality control in water quality sampling strategies	synthesis of existing literature	N/A
101 Geodynamic modelling: How to design, carry out, and interpret numerical studies	Geodynamic modelling applied to the solid Earth	Best practices for geodynamic modelling	code verification, model validation, internal consistency checks, software and data management	Code verification, model validation, data management	synthesis of existing literature, best practices	software and data management
101 geodynamic modelling: how to design, interpret, and communicate numerical studies of the solid Earth	Geodynamic modeling of the solid Earth	Providing best practices for reproducibility	Code verification, model validation, internal consistency checks, software/ data management.	Code verification, model validation, internal consistency checks, software and data management.	Synthesis of existing literature, best practices, examples	Highlights best practices for code verification, data management

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Reproducibility in Subsurface Geoscience	subsurface geoscience	Assessing reproducibility in subsurface geoscience	build a framework and minimum standards	data access, undocumented methodologies, confidentiality issues	survey of 346 Earth scientists	undocumented methodologies identified as major barriers

Evidence

(Medalie, 2020)

[Considerations for incorporating quality control into water quality sampling strategies for the U.S. Geological Survey]

> *"Reproducibility of findings is largely addressed by adequate documentation of methods and interpretations, along with accessible information. Documentation is the foundation of QA/QC. Tools and techniques that ensure robust and efficient production of environmental data are of little use without adequate documentation. The fundamental science practices at the USGS for scientific-data management require scientists to place all reviewed and interpreted datasets, along with a complete metadata record, in a permanent, trusted repository (U.S. Geological Survey, 2017)."*

(Zelst et al., 2021)

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Reproducibility in Subsurface Geoscience

> *"detailed documentation of methods and conceptual uncertainty would be beneficial to enhancing reproducibility efforts. This does not necessarily mean lengthening the paper, but instead providing adequate and detailed Supplementary Material for readers who are aiming to reproduce work"*

> *"calling for further transparency of methods employed"*

> *"implementing some form of minimum standards framework for published work"*

> *"auditable data inputs, quality control and assurance, internal and external peer-assists, tested workflows and processes, and reanalysis of previous data are commonplace-essentially a business focused system following many of the requirements for reproducibility"*

Data and Software Accessibility Standards

Modern geoscience research requires that data, software, and computational workflows be made openly accessible through persistent repositories with complete metadata and clear documentation. Following the PLUS principles (Persistent, Linked, User-friendly, Sustainable) ensures research products can be discovered, understood, and reused by the scientific community. (3 sources)

Data and software accessibility has become a cornerstone of reproducible geoscience research, requiring systematic approaches to sharing research products. The Geoscience Papers of the Future (GPF) framework emphasizes making data, software, and methods openly accessible, citable, and well documented to enable broader access by scientists, students, decision makers, and the public (Gil et al., 2016). This approach treats research products as valuable scientific assets that can accelerate scientific discovery by improving the ability of others to build upon published work (Gil et al., 2016).

Current journal practices often fall short of supporting reproducibility, as many do not require metadata that documents connections between data, software, and results (Chen et al., 2020). Geoscience data requires rich contextual information including scientific motivations for data collection, instruments used for data acquisition, pre-processing analyses performed, and scientific results obtained (Chen et al., 2020). Best practices require reusable data, software, and computational provenance published in public repositories with necessary metadata, licenses, and unique persistent identifiers for citation (Chen et al., 2020) (Gil et al., 2016).

The PLUS guidelines provide a practical framework for accessibility standards (Yu et al., 2016). **Persistent** identification requires uploading data and software to online repositories with DOIs or PURLs, while authors should register ORCID identifiers for ongoing research updates (Yu et al., 2016). **Linked** documentation ensures data and software are connected within computational workflows, including intermediate data that represent essential information in final figures (Yu et al., 2016). **User-friendly** presentation requires comprehensive metadata and software packaging with documentation and instructions that help readers determine applicability and usage, considering diverse disciplinary backgrounds (Yu et al., 2016). **Sustainable** practices involve maintaining software at repositories and providing ongoing support for research products (Yu et al., 2016) (Gil et al., 2016).

Evidence

(Gil et al., 2016)

Toward the Geoscience Paper of the Future: Best practices for documenting and sharing research from data to software to provenance

> *"This article proposes best practices for GPF authors to make data, software, and methods openly accessible, citable, and well documented. The publication of digital objects empowers scientists to manage their research products as valuable scientific assets in an open and transparent way that enables broader access by other scientists, students, decision makers, and the public. Improving documentation and dissemination of research will accelerate the pace of scientific discovery by improving the ability of others to build upon published work."*

(Chen et al., 2020)

Practices, Challenges, and Prospects of Big Data Curation: a Case Study in Geoscience

> *"journals often do not require metadata entailing the connections between the data, software, and results for reproducible and transparent research. Geoscience data by nature has a rich background and requires necessary summary and descriptions of the context, ranging from the scientifc motivations for collecting the data, the instruments for acquiring the data, the pre-processing analyses of the data, and the scientifc results (Ebert-Uphoff et al., 2017)"*

> *"(Gil et al., 2016) emphasized the importance of open science (data and software), reproducibility, and modern digital scholarship for Geoscience Papers of the Future (GPF). The suggested best practices for GPF are reusable data, software, and computational provenance of results through publication in a public repository with necessary metadata, license, and unique and persistent identifer for citation"*

(Yu et al., 2016)

Open science in practice: Learning integrated modeling of coupled surface-subsurface flow processes from scratch

> *"The GPF activity proposed best practices to make sure that geoscience papers of the future would capture the core concept of reproducibility (Gil et al., 2016). These best practices consist of 20 recommendations focusing on data accessibility, data documentation, software accessibility, software documentation, provenance documentation, method documentation, and author identification"*

> *"we summarize the PLUS guidelines for geoscience papers to make the software and data persistent, linked, user-friendly, and sustainable"*

> *"Persistent. Data, software, and authors should be persistently (i.e., consistently) identifiable"*

> *"data and software can be uploaded to online repositories with DOIs or PURLs (Permanent Uniform Resource Identifiers). Linked. Data and software should be linked in the computational workflow so that the software can be understood and reused by the readers. These links should include any intermediate data derived from the original data that represent essential information in the final figures of the article. User-friendly. To make software user-friendly is useful for the discovery and reuse among readers. Metadata are always good for the users of the data and software. The software should be packaged with documentation and*

instructions so that readers can first decide if the software can be used in their work, and how to apply it. It is always good to be thoughtful for any possible readership, especially those from other disciplines. Sustainable. Authors are recommended to register an ORCID (Open Researcher and Contributor ID), so that readers can always obtain the research updates. Software should be maintained at repositories"

Technical Implementation Details

Successful implementation of reproducible geoscience methods requires specific technical documentation including computational environment details, code verification procedures, runtime specifications, and explicit statements of all model assumptions and limitations. (3 sources)

Technical implementation of reproducible geoscience research requires systematic attention to computational and procedural details that enable others to replicate studies. Key technical requirements include:

• **Computational environment specifications**: Document where simulations were performed, number of cores used, average runtime of simulations, and whether results can be reproduced on desktop computers or require cluster access (Zelst et al., 2021) (Zelst et al., 2022)

• **Code verification and availability**: Provide code availability statements, code verification details, input files, installation guides, and other data necessary to reproduce simulation results (Zelst et al., 2021) (Zelst et al., 2022)

• **Analysis and visualization documentation**: Explicitly describe visualization software used (such as ParaView or Matlab), post-processing scripts developed in specific programming languages, and statistical methods employed for data analysis (Zelst et al., 2022)

• **Post-processing script availability**: Provide post-processing scripts for peer review and community engagement, with more detailed descriptions for complex analytical methods (Zelst et al., 2022)

• **Explicit model assumptions**: State all model assumptions clearly and explicitly in either the theoretical description or numerical approach sections, as these assumptions may not be obvious to readers but are essential for proper interpretation (Zelst et al., 2021) (Zelst et al., 2022)

• **Materials and protocol documentation**: Describe materials used in the study, explain how materials were prepared, detail the research protocol, specify how measurements were made and calculations performed, and state which statistical tests were used for data analysis (Kallet, 2004)

• **Rationale for methodological choices**: Justify the experimental design and explain why specific experimental procedures were chosen to answer the research question (Kallet, 2004)

Paper	Scope Of Best Practices	Target Audience	Focus On Reproducibility And Transparency	Inclusion Of Technical Implementation Details	Domain Specificity	Rationale For Method Choices

Paper	Scope Of Best Practices	Target Audience	Focus On Reproducibility And Transparency	Inclusion Of Technical Implementation Details	Domain Specificity	Rationale For Method Choices
101 Geodynamic modelling: How to design, carry out, and interpret numerical studies	Entire modelling process from design to communication.	first-time and experienced modellers, collaborators, and reviewers	Highlights best practices including code verification, data management.	Code verification, model validation, software/ data management	Geodynamic modelling applied to the solid Earth	Choice of governing equations to numerical methods
101 geodynamic modelling: how to design, interpret, and communicate numerical studies of the solid Earth	Entire modeling process from design to communication.	First-time and experienced modellers, collaborators, reviewers	Discusses best practices including code verification and model validation	Yes, including code verification, model validation, and data management.	Geodynamic modelling of the solid Earth	Highlights best practices including verification, validation, consistency checks
How to Write the Methods Section of a Research Paper	Method section writing best practices	Researchers	clear and precise description of experiment for validity	describes research protocol and statistical tests	General scientific research	Emphasizes the rationale for chosen procedures.

Evidence

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"The methods section is considered one of the most important parts of any scientific manuscript (Kallet, 2004). A good methods section allows other scientists to verify results and conclusions, understand whether the design of the experiment is relevant for the scientific question (validity), and to build on the work presented (reproducibility and replicability) by assessing alternative methods that might produce differing results. Thus, the major goals of the methods section are to verify the experiment layout and allow others to reproduce the results"

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"The 'who, what, when, where, how, and why' order proposed by Annesley (2010) breaks the methods section down in the following questions: who performed the experiment? What was done to answer the research question? When and where was the experiment undertaken? How was the experiment done, and how were the results analysed? Why were specific procedures chosen?"

"The methods should start with a brief outline (one paragraph) describing the study design and the main steps taken to answer the scientific question posed in the introduction. The outline should be logical and go from theoretical elements, to numerical aspects,

to analysis and post-processing. First to be described is the theoretical framework. This includes the mathematical and physical concepts used in the study including the governing equations, constitutive equations, initial and boundary conditions, and any other relevant theory describing the model setup"

"After the model setup has been explained, the methods should contain a section describing the design or layout of the study in detail. What is being tested/varied? How many simulations were performed in terms of model and parameter space?"

"Information should also be given on code and data availability"

"The authors should indicate code availability, code verification, and input files or other data necessary to reproduce the simulation results (e.g., installation guides). Additional questions to be answered in the methods section are: where were the simulations performed, and on how many cores? What was the average runtime of a simulation? Can the model results be reproduced on a laptop/desktop computer or is access to a cluster required?"

"model assumptions need to be stated clearly, in either the description of the theory or the numerical approach"

"As long as assumptions are explicit and consistent (i.e., through clear and honest communication), the reviewers and readers will find fewer flaws about the study."

(Zelst et al., 2022)

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"Analysis, visualisation and post-processing techniques of numerical data should also be described in the methods section. This is a step generally ignored, but it is important to be open about it, e.g., 'visualisation was performed in ParaView/Matlab, and post-processing scripts were developed in python/matlab/unicorn language by the author'. If the post-processing methods are more complex, the author can provide more details (i.e., statistical methods used for data analysis). It is also good practice to provide these post-processing scripts for peer-reviewing and community engagement"

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"The authors should indicate code availability, code verification, and input files or other data necessary to reproduce the simulation results (e.g., installation guides). Additional questions to be answered in the methods section are: where were the simulations performed, and on how many cores? What was the average runtime of a simulation? Can the model results be reproduced on a laptop/desktop computer or is access to a cluster required?"

"model assumptions need to be stated clearly, in either the description of the theory or the numerical approach"

"It is important to bear in mind that this set of assumptions may not always be obvious to the reader. As long as assumptions are explicit and consistent (i.e., through clear and honest communication), the reviewers and readers will find fewer flaws in the study."

(Kallet, 2004)

How to Write the Methods Section of a Research Paper

"The methods section of a research paper provides the information by which a study's validity is judged. Therefore, it requires a clear and precise description of how an experiment was done, and the rationale for why specific experimental procedures were chosen. The methods section should describe what was done to answer the research question, describe how it was done, justify the experimental design, and explain how the results were analyzed. Scientific writing is direct and orderly. Therefore, the methods section structure should: describe the materials used in the study, explain how the materials were prepared for the study, describe the research protocol, explain how measurements were made and what calculations were performed, and state which statistical tests were done to analyze the data. Once all elements of the methods section are written, subsequent drafts should focus on how to present those elements as clearly and logically as possibly. The description of preparations, measurements, and the protocol should be organized chronologically. For clarity, when a large amount of detail must be presented, information should be presented in sub-sections according to topic. Material in each section should be organized by topic from most to least important."